

Independent Quantitative Investment Research

Portfolio Optimization and Performance Evaluation of Selected Johannesburg Stock Exchange Equities

A Quantitative Investment Analysis

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Keywords

Portfolio Optimization • Modern Portfolio Theory • Efficient Frontier • Sharpe Ratio
• Backtesting • Johannesburg Stock Exchange

0.1 Abstract

Portfolio optimization remains one of the fundamental challenges in quantitative investment management, requiring the construction of portfolios that maximize expected returns while controlling investment risk. This report presents the development and evaluation of a quantitative portfolio optimization framework using historical equity data from selected companies listed on the Johannesburg Stock Exchange (JSE).

Historical daily adjusted closing prices were collected and preprocessed to compute asset returns and estimate the statistical inputs required for portfolio optimization. Asset expected returns, covariance matrices and correlation structures were estimated, with covariance shrinkage employed to improve the stability of risk estimates. Modern Portfolio Theory was subsequently applied to construct both the Minimum Variance Portfolio and the Maximum Sharpe Ratio Portfolio. The efficient frontier was generated through portfolio simulations to illustrate the trade-off between expected return and portfolio risk.

The optimized portfolio was evaluated through historical backtesting using cumulative returns, wealth growth, and maximum drawdown analysis. Portfolio performance was benchmarked against the Johannesburg Stock Exchange All Share Index (JSE ALSI) to assess the effectiveness of the optimization strategy under actual market conditions. Results indicate that the optimized portfolio delivered superior cumulative performance relative to the benchmark while maintaining an acceptable level of portfolio risk throughout the investment period.

The findings demonstrate the practical application of quantitative portfolio optimization techniques in equity portfolio construction and highlight the importance of disciplined, data-driven investment decision-making. Although the analysis assumes frictionless markets and static portfolio weights, the framework provides a solid foundation for more advanced portfolio management techniques incorporating dynamic rebalancing, transaction costs and alternative risk models.

Executive Summary

This report presents the construction, optimization, and performance evaluation of an equity investment portfolio comprising selected companies listed on the Johannesburg Stock Exchange (JSE). The primary objective was to investigate whether quantitative portfolio optimization techniques could produce a portfolio that delivers superior risk-adjusted performance relative to the Johannesburg Stock Exchange All Share Index (JSE ALSI).

Historical daily price data were collected for seventeen large-cap JSE-listed companies and processed to estimate expected returns, portfolio risk, and asset correlations. Modern Portfolio Theory (MPT) served as the foundation for portfolio construction, with optimization performed under both the Minimum Variance and Maximum Sharpe Ratio criteria. To improve the robustness of portfolio risk estimates, covariance shrinkage techniques were incorporated into the optimization framework.

The optimized portfolio was subsequently evaluated using historical backtesting techniques. Performance was assessed through cumulative returns, wealth growth, portfolio drawdown, and comparison against the JSE ALSI benchmark. The empirical results indicate that the optimized portfolio achieved stronger cumulative performance than the benchmark while maintaining an acceptable level of downside risk over the investment horizon.

Several graphical analyses were produced to support the investment findings, including the efficient frontier, asset correlation heatmap, portfolio allocation charts, drawdown analysis, and benchmark performance comparison. These visualizations provide insight into the diversification characteristics, risk-return trade-off, and overall effectiveness of the optimization strategy.

Overall, the results demonstrate that quantitative portfolio optimization provides a structured and data-driven approach to equity portfolio construction. Although the analysis assumes static portfolio weights and excludes practical considerations such as transaction costs and periodic rebalancing, the framework establishes a strong foundation for

more advanced portfolio management applications. Future extensions may incorporate dynamic optimization, alternative risk measures, and machine learning techniques to further enhance portfolio performance.

Contents

0.1 Abstract	i
Contents	iv
List of Figures	v
List of Tables	vi
0.2 Background	1
0.3 Problem Statement	2
0.4 Objectives	2
0.5 Scope of the Study	3
0.6 Structure of the Report	3
Bibliography	4

List of Figures

List of Tables

Introduction1

0.2 Background

Portfolio management is fundamentally concerned with allocating capital across a set of financial assets in a manner that achieves an appropriate balance between expected return and investment risk. As financial markets have become increasingly complex and data-driven, quantitative methods have emerged as indispensable tools for supporting objective investment decisions. Rather than relying solely on intuition or subjective judgment, quantitative portfolio management applies statistical modelling, mathematical optimization, and financial theory to construct portfolios that seek to maximise returns while controlling risk.

Modern Portfolio Theory (MPT), introduced by Harry Markowitz (1952), remains one of the most influential frameworks in investment management. The theory demonstrates that portfolio risk depends not only on the individual risk of each asset but also on the relationships between asset returns. Consequently, investors can improve portfolio efficiency through diversification by combining assets whose returns are imperfectly correlated. This principle forms the foundation of many contemporary portfolio construction techniques employed by investment managers worldwide.

The Johannesburg Stock Exchange (JSE) represents the largest and most liquid securities exchange in Africa, providing investors with exposure to companies operating across multiple sectors of the South African economy. Given its economic significance and sectoral diversity, the JSE provides an appropriate environment for evaluating quantitative portfolio optimization techniques and assessing their practical investment performance.

This report develops a quantitative portfolio optimization framework using historical equity data from selected JSE-listed companies. The analysis combines statistical estimation, Modern Portfolio Theory, covariance shrinkage, and historical backtesting to construct and evaluate an optimized investment portfolio. Portfolio performance is subsequently benchmarked against the Johannesburg Stock Exchange All Share Index (JSE

ALSI) to determine whether optimization improves investment outcomes relative to a passive market investment.

0.3 Problem Statement

Although diversification remains a fundamental principle of investment management, constructing an efficient portfolio becomes increasingly difficult as the number of available securities expands. Investors must simultaneously estimate expected returns, portfolio risk, and the relationships among asset returns while balancing conflicting objectives of maximizing returns and minimizing risk. Traditional portfolio construction approaches based on subjective judgment may therefore produce allocations that are suboptimal or insufficiently diversified.

Furthermore, estimation error in expected returns and covariance matrices may significantly affect portfolio optimization results. Consequently, there is a need for robust quantitative techniques capable of producing well-diversified portfolios while maintaining favourable risk-adjusted performance under realistic market conditions.

0.4 Objectives

The primary objective of this study is to develop and evaluate a quantitative portfolio optimization framework for selected Johannesburg Stock Exchange equities.

The specific objectives are:

1. Collect and preprocess historical daily price data for selected JSE-listed companies.
2. Estimate expected asset returns and covariance matrices required for portfolio optimization.
3. Apply covariance shrinkage techniques to improve portfolio risk estimation.
4. Construct Minimum Variance and Maximum Sharpe Ratio portfolios using Modern Portfolio Theory.
5. Evaluate portfolio performance through historical backtesting.
6. Compare the optimized portfolio against the Johannesburg Stock Exchange All Share Index benchmark.

0.5 Scope of the Study

The analysis focuses exclusively on a portfolio of selected equities listed on the Johannesburg Stock Exchange. Historical daily adjusted closing prices are used to estimate returns and portfolio risk. Portfolio optimization is performed under the assumptions of Modern Portfolio Theory using historical estimates of expected returns and covariance matrices.

The study evaluates portfolio performance using cumulative returns, wealth growth, drawdown analysis, and benchmark comparison. Practical market considerations such as transaction costs, taxes, liquidity constraints, and dynamic portfolio rebalancing are excluded from the analysis.

0.6 Structure of the Report

The remainder of this report is organized as follows.

Chapter 2 describes the dataset, investment universe, and preprocessing procedures. Chapter 3 presents the portfolio optimization methodology, including return estimation, covariance estimation, covariance shrinkage, and optimization techniques. Chapter 4 presents the empirical findings and discusses the optimization results. Chapter 5 evaluates portfolio performance through historical backtesting and benchmark comparison. Finally, Chapter 6 summarizes the principal findings, discusses limitations, and provides recommendations for future research.

Bibliography